

An Augmented Reality Platform for Medical Self-Diagnosis Training

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ABSTRACT

Diagnosing medical conditions is critical for medical personnel. Current training tools include textbooks or descriptive scenarios that lack interactivity and offer limited hands-on experience or expensive scenarios with actors. Interactive virtual simulations hold great promise, but are limited by a lack of tangibility and realism. We suggest that augmented reality may be a key addition to the training toolbox, specifically augmenting the user's body, thus providing a tangible platform for interaction. We developed such a self-diagnosis training platform and performed basic usability testing. We found that users could successfully use the platform, with performance correlated with self-reported medical knowledge.

Index Terms: Medical Simulation, Diagnosis, Medical Training, Full Body Tracking, Augmented Reality

1 INTRODUCTION

Accurate diagnosis forms the foundation of effective medical treatment. However, traditional medical training often relies on passive learning methods, such as textbooks and lectures, which lack the interactivity and realism necessary to effectively cultivate diagnostic skills [1]. Real clinic experience or simulated scenarios with actors can overcome these challenges; however, these options are limited by cost and by the availability of the instructor or actor [1,2]. Consequently, there is a pressing need for innovative training tools that can bridge the gap between theoretical knowledge and practical application. Virtual Reality (VR) has emerged as a promising avenue for enhancing medical training. VR provides an immersive and interactive environment, as well as an active learning experience [3-5]. In previous research VR has been proven to enhance the motivation and the agency during training [3,6]. However, VR has inherent limitations: it disconnects users from their surroundings, potentially causing safety issues, it requires the creation of a realistic environment which can be costly, and interactions with virtual objects lack tangibility which can diminish realism. Augmented reality (AR) offers the potential for mitigating these challenges by overlaying virtual content onto the real world. This has the advantages offered by VR's flexibility while preserving the user's connection to their physical surroundings [7,8]. Additionally, by placing the augmented content on a tangible object such as a target manikin or the user's own body users can interact tangibly with the simulation, allowing interactions like placing a bandage on a wound. Utilizing the user's own body also increases interactivity vs. a static manikin (e.g. users can rotate their hands to see a medical condition from different angles). Thus, here we designed and implemented an AR platform for medical self-diagnosis training and evaluation. This platform allows users to see virtual representations of different medical conditions on their own bodies. It was designed modularly

to enable the integration of new conditions and to easily upgrade the graphics on existing ones. Our main focus here is to implement our device and to evaluate the prototype's usability.

2 METHODS

2.1 Participants

21 healthy volunteers (11 females) participated in our study. The study was approved by BGU's IRB and all participants provided informed consent, and received no monetary compensation.

2.2 Hardware

We used the Meta Quest 3 head-mounted display which offers passthrough AR and has a built-in upper body tracking system. Our prototype is hand controlled (i.e. no controllers needed).

2.3 Software

We designed the functionality and user interface (UI) of our application using the Unity3D platform and the Meta XR SDK. We designed the virtual representations of symptoms using Blender and a full body model from Mixamo.

2.4 The Tool

When entering the tool the user is offered a menu with options of different diseases. When an option is chosen, the specific disease is overlaid on the users' arm/hand and moves with it naturally. The user continues to view their regular surroundings via passthrough augmented reality. The tool has a testing mode in which the user experiences an augmented disease on themselves and then need to select from a 5-choice menu which disease they think it is (Fig. 1(d)).

2.5 Usability Experiment Paradigm

The participants received a short explanation about the experiment and software, and provided their informed consent. They then donned the headset and experienced our tool via guided instructions from the experimenter, and were then tasked to identify three conditions (snake bite, rash, dermatophytosis; order randomized) from a list with 5 options. No feedback was given after each answer. Note that participants were not trained on these symptoms and their recognition relied upon their prior knowledge with the condition – this enabled us to test for very basic face and construct validity as they would only recognize the symptoms above chance if we had simulated them well enough, and we would expect to see a difference between participants with medical knowledge and those who did not have one. Finally participants filled out a questionnaire.

2.6 Experimental Measures

As this was only the basic usability test, we tested functionality (i.e. could users correctly use the tool, assessed via experimenter's log and via questionnaire), participants subjective feedback (e.g. was it easy to use? Immersive? Enjoyable? and if they experienced any discomfort while using it, assessed via questionnaire), participants self-professed general medical knowledge, and participants' answer accuracy (recorded automatically from within the tool).

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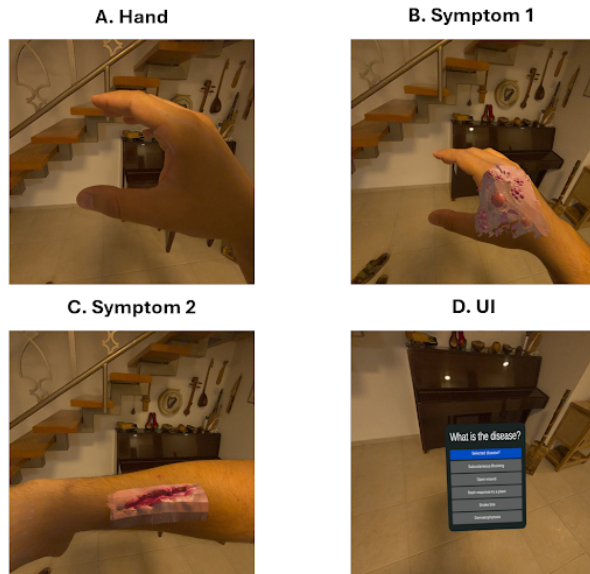


Figure 1: Visuals of the application. (A) Image of subject's hand without augmentation. (B) Image of the subject's hand with a rash representation on it. (C) Image of subject's hand with a representation of an open wound. (D) The UI system for diagnosing the disease.

3 RESULTS

3.1 Overall

All participants were easily able to complete the task, and no functionality problems or adverse effects were noted by the experimenters.

3.2 Subjective Experience

All the subjective results were scored on a Likert scale between 1 to 5 where 1 is the lowest and 5 highest. These values are represented in the following form: Mean $\pm$ SE. We interpret median scores above 3 as positive and below 3 as negative.

Presence: Participants reported a presence level of 4.3 $\pm$ 0.2.

Comfort: Participants reported a comfort level of 4.5 $\pm$ 0.16. and a UI usability level of 4.1 $\pm$ 0.2.

Realism: Participants reported a natural movement score of 4.1 $\pm$ 0.19 and realism level of the symptoms as 3.2 $\pm$ 0.2.

Enjoyment: Participants reported an enjoyment level of 4.6 $\pm$ 0.14.

Usefulness: All 21 of the participants reported that they believe such a system could assist them in studying medical diagnosis.

3.3 Objective Performance

Participants were generally able to correctly recognize the condition from among the 5 options (chance level at 20%), with an overall performance of 53% $\pm$ 7% (Rank-sum test, $p < 0.05$). Breaking down performance by condition we found that all three of the conditions – rash, snake bite and fungal infection – were significantly better than chance (47% $\pm$ 11%, 47% $\pm$ 11%, 66% $\pm$ 10% respectively, Rank-sum test, all $p < 0.05$).

3.4 Performance and Subjective Knowledge

We found a significant correlation ($\rho = 0.44$, $p < 0.05$) between participants' self-reported level of medical knowledge and their performance in our task.

4 DISCUSSION AND FUTURE WORK

We successfully implemented our platform and all participants could successfully use it, correctly answering the questions with positive scores and in a manner correlating to their self-professed medical knowledge. Subjectively, participants found our platform enjoyable, induced presence, comfortable and easy to use. They found the movement to be natural, but found the graphics only somewhat realistic. Thus, on a basic usability level our platform has achieved its benchmarks. However, this is but a first step towards the practical use of such a platform as the current study was limited to usability aspects. We are currently further developing it by improving graphics, adding a long list of other medical conditions and further improving the user interface. The use of the users own arm enables a tangible scaffolding, and future work will utilize this to practice treatment - e.g. proper bandaging over a wound. Finally, We plan to run a wider study with medical professionals and with a fully powered group of students to test participants' ability to diagnose these symptoms before and after training on them in their medical training course during spring semester 2025.

5 CONCLUSION

In conclusion, we successfully implemented an AR platform which projects augmented symptoms on the user's body and successfully tested its usability, setting the stage for deploying and testing it in the field with students and medical professionals.

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REFERENCES

- [1] H. Yang et al., "Virtual Standardized Patients Versus Traditional Academic Training for Improving Clinical Competence Among Traditional Chinese Medicine Students: Prospective Randomized Controlled Trial," *Journal of Medical Internet Research*, vol. 25, no. 1, p. e43763, Sep. 2023, doi: 10.2196/43763.
- [2] A. Juhi, M. J. Pinjar, G. Marndi, B. R. Hungund, and H. Mondal, "Evaluation of Blended Learning Method Versus Traditional Learning Method of Clinical Examination Skills in Physiology Among Undergraduate Medical Students in an Indian Medical College," *Cureus*, vol. 15, no. 4, p. e37886, Apr. 2023, doi: 10.7759/cureus.37886.
- [3] N. Cevallos, B. Zukotynski, D. Greig, M. Silva, and R. M. Thompson, "The Utility of Virtual Reality in Orthopedic Surgical Training," *Journal of Surgical Education*, vol. 79, no. 6, pp. 1516–1525, Nov. 2022, doi: 10.1016/j.jsurg.2022.06.007.
- [4] J. Polson, G. Barrie, R. J. Ribeiro, and K. V. Sarma, "Implementation of a Proof-of-Concept Virtual Reality Medical Simulation Training Capability for USAF Pararescuemen," 2023.
- [5] S. Savir et al., "Virtual Reality: The Future of Invasive Procedure Training?," *Journal of Cardiothoracic and Vascular Anesthesia*, vol. 37, no. 10, pp. 2090–2097, Oct. 2023, doi: 10.1053/j.jvca.2023.06.032.
- [6] M. U. Sattar, S. Palaniappan, A. Lokman, N. Shah, U. Khalid, and R. Hasan, "Motivating Medical Students Using Virtual Reality Based Education," *International Journal of Emerging Technologies in Learning (iJET)*, vol. 15, no. 02, Art. no. 02, Jan. 2020, doi: 10.3991/ijet.v15i02.11394.
- [7] J. S. Ruthberg et al., "Mixed reality as a time-efficient alternative to cadaveric dissection," *Medical Teacher*, Aug. 2020, [Online]. Available: <https://www.tandfonline.com/doi/abs/10.1080/0142159X.2020.1762032>
- [8] G. Baratz, P. S. Sridharan, V. Yong, C. Tatsuoaka, M. A. Griswold, and S. Wish-Baratz, "Comparing learning retention in medical students using mixed-reality to supplement dissection: a preliminary study," *International Journal of Medical Education*, vol. 13, p. 107, Apr. 2022, doi: 10.5116/ijme.6250.0af8.